Online Appendix 2
Sensitivity Analysis of Alternative Fixed Hybrid Forecast-Combination Weights

This appendix reports an additional robustness analysis examining whether the Hybrid model's forecasting performance is sensitive to alternative fixed ARIMAX–LSTM forecast-combination weights. In the main analysis, the Hybrid model uses OLS-based stacking, where the forecast-combination weights are estimated from the training sample and then applied unchanged to the validation and test periods. To assess robustness, we compare this data-driven approach with alternative fixed weighting schemes: 0.3/0.7, 0.4/0.6, 0.5/0.5, 0.6/0.4, and 0.7/0.3 ARIMAX–LSTM combinations.

Table A2.1. Alternative Fixed ARIMAX–LSTM Hybrid Weighting Schemes

Model	ARIMAX Weight	LSTM Weight	Description
Hybrid 1	0.3	0.7	Higher LSTM contribution
Hybrid 2	0.4	0.6	Moderate LSTM dominance
Hybrid 3	0.5	0.5	Equal-weight Hybrid
Hybrid 4	0.6	0.4	Moderate ARIMAX dominance
Hybrid 5	0.7	0.3	Higher ARIMAX contribution

Table A2.2. Full-Period Forecast Performance under Alternative Fixed Hybrid Weights

ARIMAX Weight	LSTM Weight	ARIMAX RMSE	LSTM RMSE	Hybrid RMSE	ARIMAX MAE	LSTM MAE	Hybrid MAE	ARIMAX R^2	LSTM R^2	Hybrid R^2	Improvement vs ARIMAX (%)	Improvement vs LSTM (%)
0.3	0.7	0.055837	0.013728	0.021960	0.045530	0.009356	0.017789	-16.086392	-0.032825	-1.642783	60.67	-59.96
0.4	0.6	0.055837	0.013426	0.025594	0.045530	0.008849	0.020572	-16.086392	0.012188	-2.589792	54.16	-90.63
0.5	0.5	0.055837	0.013451	0.030316	0.045530	0.008861	0.024558	-16.086392	0.008400	-4.036709	45.71	-125.37
0.6	0.4	0.055837	0.013481	0.035534	0.045530	0.008996	0.028864	-16.086392	0.004072	-5.919935	36.36	-163.59
0.7	0.3	0.055837	0.013524	0.040406	0.045530	0.008989	0.032829	-16.086392	-0.002330	-7.947372	27.64	-198.77

Notes: *This table reports full-period out-of-sample forecast performance under alternative fixed ARIMAX–LSTM Hybrid weights. Lower RMSE and MAE values indicate better predictive accuracy. The improvement columns measure the percentage reduction in RMSE of the Hybrid model relative to ARIMAX and LSTM, respectively. Negative improvement relative to LSTM indicates that the standalone LSTM performs better than the fixed-weight Hybrid model.*

Table A2.3. Average Crisis-Period Forecast Performance under Alternative Fixed Hybrid Weights

ARIMAX Weight	LSTM Weight	Avg. ARIMAX RMSE	Avg. LSTM RMSE	Avg. Hybrid RMSE	Avg. ARIMAX MAE	Avg. LSTM MAE	Avg. Hybrid MAE	Avg. ARIMAX R^2	Avg. LSTM R^2	Avg. Hybrid R^2	Avg. Improvement vs ARIMAX (%)	Avg. Improvement vs LSTM (%)
0.3	0.7	0.036616	0.017037	0.019830	0.030679	0.012948	0.015844	-8.384918	-0.247078	-0.906016	35.39	-23.56
0.4	0.6	0.036616	0.016911	0.021827	0.030679	0.012763	0.017695	-8.384918	-0.253735	-1.548852	32.07	-38.18
0.5	0.5	0.036616	0.017961	0.024402	0.030679	0.013845	0.020175	-8.384918	-0.530872	-2.312447	24.64	-52.59
0.6	0.4	0.036616	0.017797	0.026356	0.030679	0.013767	0.021770	-8.384918	-0.319162	-3.148424	22.84	-67.44
0.7	0.3	0.036616	0.016340	0.028526	0.030679	0.012169	0.023597	-8.384918	-0.164031	-4.198910	18.61	-89.14

Notes: *This table reports the average performance across the crisis episodes considered in the paper: the Global Financial Crisis, Eurozone sovereign debt crisis, oil price collapse, COVID-19 shock, inflation tightening, Russia–Ukraine war, and Israel–Palestine conflict. The results show that higher LSTM weights generally improve the fixed-weight Hybrid model's crisis-period performance.*

Table A2.4. Crisis-Period Hybrid RMSE across Alternative Fixed Weighting Schemes

Crisis Period	ARIMAX RMSE	Best LSTM RMSE	Hybrid 0.3/0.7	Hybrid 0.4/0.6	Hybrid 0.5/0.5	Hybrid 0.6/0.4	Hybrid 0.7/0.3	Best Fixed-Weight Outcome
Global Financial Crisis	0.030414	0.025659	0.028427	0.027521	0.027507	0.030564	0.028467	LSTM
Eurozone Debt Crisis	0.014214	0.015043	0.016164	0.014479	0.017448	0.013878	0.013713	Hybrid 0.7/0.3
Oil Price Collapse	0.016346	0.010950	0.010716	0.012337	0.012887	0.012573	0.013352	Hybrid 0.3/0.7
COVID-19 Shock	0.054714	0.022601	0.027427	0.030180	0.033931	0.037558	0.041498	LSTM
Inflation Tightening	0.075081	0.015243	0.025898	0.033596	0.040452	0.045818	0.053311	LSTM
Russia–Ukraine War	0.028007	0.015692	0.016034	0.016982	0.018499	0.019756	0.021722	LSTM
Israel–Palestine Conflict	0.037533	0.007321	0.014145	0.017692	0.020086	0.024347	0.027622	LSTM

Notes: *This table compares RMSE values across crisis periods. Among the fixed-weight Hybrid specifications, the 0.3/0.7 ARIMAX–LSTM configuration performs best overall, particularly during the oil price collapse. However, the standalone LSTM remains strongest in several abrupt or highly nonlinear shock periods, including COVID-19, inflation tightening, the Russia–Ukraine war, and the Israel–Palestine conflict. The Eurozone debt crisis is the only case where a higher ARIMAX contribution produces the best fixed-weight Hybrid result.*

Table A2.5. Full-Period Statistical Tests under Alternative Fixed Hybrid Weights

ARIMAX Weight	LSTM Weight	DM: ARIMAX vs Hybrid	DM: LSTM vs Hybrid	MSE-F: ARIMAX vs Hybrid	MSE-F: LSTM vs Hybrid	ENC-NEW: ARIMAX vs Hybrid	ENC-NEW: LSTM vs Hybrid
0.3	0.7	25.7702	-14.4863	6853.4887	-763.9250	26.8503	-2.4115
0.4	0.6	26.4563	-16.5970	4714.6843	-908.9337	27.4869	-0.3455
0.5	0.5	26.5770	-18.9251	3000.0351	-1007.1193	27.3111	-0.2671
0.6	0.4	26.7823	-21.0776	1842.3203	-1073.5223	27.3148	-1.7406
0.7	0.3	26.8270	-22.1569	1140.7073	-1113.5206	27.1894	-0.9415

Notes: *This table reports full-period statistical tests comparing ARIMAX and LSTM against the fixed-weight Hybrid forecasts. The Diebold–Mariano statistics indicate that the Hybrid model significantly improves over ARIMAX across all fixed-weight combinations. However, the negative LSTM vs Hybrid statistics indicate that the standalone LSTM remains statistically superior to the fixed-weight Hybrid model over the full test period. This supports the main paper's conclusion that the LSTM performs strongly in the full sample, while Hybrid models become more relevant under regime-dependent conditions.*

Table A2.6. Best-Model Counts across Crisis Periods under Alternative Fixed Hybrid Weights

ARIMAX Weight	LSTM Weight	ARIMAX Wins	LSTM Wins	Hybrid Wins
0.3	0.7	1	4	2
0.4	0.6	1	5	1
0.5	0.5	1	5	1
0.6	0.4	1	5	1
0.7	0.3	0	6	1

Notes: *Best-model counts are based on RMSE performance across the crisis periods. The 0.3/0.7 ARIMAX–LSTM configuration produces the highest number of Hybrid wins among the tested fixed-weight schemes. However, the LSTM remains the most frequent best-performing model across crisis episodes, confirming the importance of nonlinear dynamics in short-horizon return forecasting.*

Table A2.7. Detailed Crisis-Period Performance for the Best Fixed-Weight Hybrid Model: ARIMAX 0.3 / LSTM 0.7

Period	ARIMAX RMSE	LSTM RMSE	Hybrid RMSE	ARIMAX MAE	LSTM MAE	Hybrid MAE	ARIMAX R^2	LSTM R^2	Hybrid R^2	Improvement vs ARIMAX (%)	Improvement vs LSTM (%)
Full Period	0.055837	0.013728	0.021960	0.045530	0.009356	0.017789	-16.086392	-0.032825	-1.642783	60.67	-59.96
Global Financial Crisis	0.030414	0.028767	0.028427	0.023515	0.022357	0.022123	-0.592992	-0.425120	-0.391689	6.53	1.18
Eurozone Debt Crisis	0.014214	0.018140	0.016164	0.011023	0.015111	0.013367	-0.289057	-1.099390	-0.666875	-13.71	10.89
Oil Price Collapse	0.016346	0.010950	0.010716	0.013411	0.008340	0.008506	-1.398428	-0.076298	-0.030762	34.44	2.14
COVID-19 Shock	0.054714	0.022854	0.027427	0.041151	0.014773	0.020025	-4.827386	-0.016727	-0.464358	49.87	-20.01
Inflation Tightening	0.075081	0.015296	0.025898	0.069171	0.011924	0.021238	-23.371620	-0.011526	-1.899657	65.51	-69.31

Russia–Ukraine War	0.028007	0.015719	0.016034	0.023308	0.012268	0.013082	-2.216569	-0.013262	-0.054168	42.75	-2.00
Israel–Palestine Conflict	0.037533	0.007532	0.014145	0.033171	0.005861	0.012569	-25.998377	-0.087224	-2.834605	62.31	-87.80

Notes: *This table reports the detailed performance of the best overall fixed-weight Hybrid configuration, based on the lowest full-period and average crisis-period RMSE among the tested fixed-weight combinations. The 0.3/0.7 ARIMAX–LSTM Hybrid model substantially improves over ARIMAX in most crisis periods, although it generally remains inferior to the standalone LSTM in abrupt or highly nonlinear shock periods.*

Discussion

The sensitivity analysis shows that the Hybrid model's performance varies across alternative fixed ARIMAX–LSTM weighting schemes. This finding confirms that forecast-combination weights matter and that arbitrarily selecting fixed weights may influence the reported performance of the Hybrid model. In the main analysis, however, the Hybrid model does not rely on manually chosen weights. Instead, it follows a data-driven OLS-stacking procedure, where the combination weights are estimated from the training sample based on the relationship between realized returns and the forecasts produced by the ARIMAX and LSTM models. These estimated weights are then fixed and applied unchanged to the validation and test periods, preserving the strictly out-of-sample design and avoiding subjective weight selection.

The alternative fixed-weight results are therefore used only as a robustness check. Across the tested combinations, increasing the LSTM contribution generally improves forecast accuracy. The best fixed-weight specification is the 0.3/0.7 ARIMAX–LSTM combination, which produces the lowest full-period Hybrid RMSE and the lowest average crisis-period Hybrid RMSE among the manually tested fixed-weight schemes. This suggests that nonlinear information captured by the LSTM contributes strongly to forecasting performance, while a modest ARIMAX component can still provide useful linear information in selected crisis regimes.

At the same time, the fixed-weight Hybrid models do not consistently outperform the standalone LSTM model, particularly during abrupt and highly nonlinear shock periods such as COVID-19, inflation tightening, the Russia–Ukraine war, and the Israel–Palestine conflict. However, they generally improve substantially over the linear ARIMAX benchmark, and in some crisis regimes, such as the oil price collapse and the Eurozone debt crisis, combining linear and nonlinear forecasts improves predictive accuracy. These findings reinforce the main conclusion that the relative performance of LSTM and Hybrid models is regime-dependent rather than uniform across all market conditions.

Overall, the robustness analysis supports the methodological choice adopted in the main manuscript. Because fixed weights can materially affect Hybrid performance, relying on OLS-based stacking is preferable to imposing arbitrary forecast-combination weights. The main Hybrid specification allows the data to determine the relative contribution of ARIMAX and LSTM during the training stage, after which the estimated weights are fixed for out-of-sample evaluation. The fixed-weight sensitivity results therefore do not replace the main Hybrid model; rather, they confirm that the main findings are not driven by a single arbitrary weighting choice and further support the use of a data-driven forecast-combination framework for evaluating CPU-related return predictability across regimes.